GreenScope Analytics – EAGC Strategic Partnership

Expanded Implementation Methodology

Pilot → Platform → Hub

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1 PHASE 1: STRATEGIC PROOF OF CONCEPT (PILOT)

1.1 Overview

Attribute	Specification
Duration	3 weeks (extendable to 4 weeks if data access delays)
Objective	Validate causal intelligence on one "impossible question" with quantified confidence
Success Threshold	≥85% causal attribution confidence, ≥50% decision speed improvement
Decision Authority	EAGC CEO (go/no-go for Phase 2)

1.2 Inputs Required from EAGC

Input Category	Specific Items	Format	Owner	Timing
Strategic	One "impossible question" selected from priority list	Written brief	EAGC CEO/Strategy	Day 1
	Corridor/country scope confirmation (e.g., Kampala-Nairobi)	Workshop output	Joint	Day 1
Data	RATIN historical price series (24+ months)	API access or CSV	RATIN Manager	Days 2–3
	Trade flow data (formal and informal estimates)	Spreadsheets, reports	Trade Analyst	Days 2–3
	Stock estimates/strategic reserve levels	Confidential database	Operations Manager	Days 2–3
	Export/import restriction records (dates, policies)	Policy documents	Policy Lead	Days 2–3
Expert Knowledge	2–3 senior economists for causal hypothesis sessions	4-hour workshop	Chief Economist	Days 4–5
	Historical context on 2–3 past market crises	Narrative debrief	Senior Analyst	Day 5
Operational	Dedicated workspace for GreenScope team (if in-person)	Office access	Administration	Week 1
	IT security clearance for external data tools	Approved vendor status	IT/Security	Days 1–2
Decision	Pilot success criteria sign-off	Signed document	EAGC CEO	Day 1
	Go/no-go decision authority	Designated executive	EAGC CEO	Day 17

1.3 Week 1: Discovery and Data Integration

Day	Activity	Deliverable	Owner
1	Kickoff workshop with EAGC	Aligned stakeholders, confirmed question scope	Joint
2–3	Data audit: RATIN access + gap assessment	Data inventory, quality scorecard	GreenScop
4–5	External data integration (climate, conflict, satellite)	Unified dataset, harmonization rules	GreenScop
5	Expert interview: EAGC economists validate causal hypotheses	Validated causal graph structure	Joint

1.3.1 Key Activities

- **Data Integration Layer Deployment:** Automated ingestion from RATIN APIs; manual upload protocols for historical gaps
- **Data Quality Scoring:** Systematic flagging of missing trade flow data, price reporting delays, stock estimation gaps
- **Causal Hypothesis Elicitation:** Structured interviews capturing domain knowledge (e.g., "Export restrictions in Uganda typically take 2–3 weeks to affect Nairobi prices")

1.3.2 Risk Mitigation

Risk	Mitigation Strategy
RATIN data incomplete	Deploy proxy variables: satellite vegetation indices for production, conflict data for route disruption
Question scope too broad	Narrow to specific corridor (e.g., Kampala-Nairobi maize flow) and time-bound historical event
Expert availability limited	Record asynchronous interviews; use written questionnaire backup

1.4 Week 2: Model Calibration and Analysis

Day	Activity	Deliverable	Owner	Success
6–8	Causal discovery algorithms on historical data	Draft causal model (DAG structure)	GreenScope	Model co
9	Counterfactual simulation setup	Scenario engine operational	GreenScope	3+ scenar
10	Uncertainty quantification and confidence scoring	Confidence intervals for all outputs	GreenScope	≥85% att
11–12	Initial findings review with EAGC team	Draft attribution report	Joint	EAGC va

1.4.1 Technical Methodology

1. **Causal Structure Learning:** PC algorithm or GES (Greedy Equivalence Search) to discover relationships from data, constrained by expert knowledge

2. **Bayesian Inference:** Estimate effect sizes with credible intervals (e.g., "Export restrictions explain 35% of price spike, CI: 28–42%")
3. **Counterfactual Simulation:** Do-calculus to estimate "what if" scenarios (e.g., "If reserves released 2 weeks earlier, price spike reduced by 12%")

1.4.2 Quality Gates

- Model achieves >85% confidence on root cause attribution
- Counterfactuals pass "smell test" with EAGC economists
- Dashboard loads in <3 seconds

1.5 Week 3: Delivery and Capability Transfer

Day	Activity	Deliverable	Owner	Success Criteri
13–14	Dashboard refinement and documentation	User guide, methodology brief	GreenScope	EAGC team navi
15	Training session: EAGC analysts	3+ trained users, competency check	Joint	Users recreate an
16	Final presentation to EAGC leadership	Board-ready findings deck	GreenScope	CEO/COO sign-
17	Pilot retrospective and go/no-go decision	Decision document, Phase 2 scope (if proceeding)	Joint	Clear next steps

1.5.1 Deliverables

1. **Working Dashboard:** Interactive exploration of causal model with scenario testing interface
2. **Methodology Report:** Documentation of analytical approach, data limitations, confidence assessments
3. **Training Materials:** Recorded tutorials, user manual, troubleshooting guide
4. **Pilot Evaluation Scorecard:** Measured performance against success criteria

1.6 Decision Gate

Outcome	Criteria	Next Step
Success	≥85% causal confidence, ≥50% decision speed improvement, team operational	Proceed to Phase 2
Partial	70–85% confidence, some improvement, refinement needed	Remediation period or mutual termination
Unsuccessful	<70% confidence, no speed improvement	Insights retained, partnership concluded

2 PHASE 2: PLATFORM PARTNERSHIP (Year 1)

2.1 Overview

Attribute	Specification
Duration	Months 2–12 (10 months active deployment)
Objective	Scale to 3-country operational platform with 4 intelligence modules
Geographic Scope	3 priority countries from: Kenya, Uganda, Tanzania, Rwanda, Burundi, South Sudan
Commodity Coverage	Maize (priority), beans, wheat, sorghum, rice (conditional on data)
Strategic Outcome	EAGC becomes regional decision leader with autonomous capabilities

2.2 Inputs Required from EAGC

Input Category	Specific Items	Format	Owner	Timing
Strategic	Country prioritization (3 from candidate list)	Signed priority list	EAGC Board/CEO	Month 2
	Commodity prioritization beyond maize	Ranked list	Chief Economist	Month 2
	Integration with EAGC strategic plan 2024–2028	Strategy document	CEO	Month 2
Data	RATIN API access (expanded to 3 countries, 5+ commodities)	API credentials, documentation	IT/RATIN Manager	Month 2–3
	Historical intervention records (reserve releases, corridor actions, outcomes)	Database, reports	Operations	Month 2–3
	Member state trade policy databases	Policy trackers, contact lists	Policy Lead	Month 3
	EAGC network contacts in 3 target countries	Introduction letters, MOU templates	Regional Coordinators	Month 3
Expert Knowledge	5+ analysts for user research and testing	Interview time, workflow observation	Department Heads	Months 2–3
	Minister/board briefing templates and preferences	Past presentations, style guides	Communications	Month 3
	Standard Operating Procedures for crisis response	SOP documents	Operations	Month 4
Operational	Dedicated EAGC counterpart (20% time commitment)	Named senior analyst/manager	HR/CEO	Month 2
	Workspace for GreenScope team (if hybrid)	Office space, meeting rooms	Administration	Ongoing
	IT integration support (firewalls, single sign-on)	IT liaison, security review	IT Director	Months 2–3

Table 2 continued from previous page

Input Category	Specific Items	Format	Owner	Timing
Partnership	Introductions to national statistics offices, customs authorities	Letters, meeting facilitation	Regional Directors	Months 3–4
	Data sharing agreement templates	Legal precedents	Legal Counsel	Month 3
Decision	Module prioritization and scope adjustments	Signed change requests	CEO/Steering Committee	Monthly
	User adoption targets and accountability	KPI agreement	Department Heads	Month 4

2.3 Months 2–3: Foundation and Expansion Planning

Week	Activity	Deliverable	Owner	Dependencies
5–6	Technical architecture scale-up	Cloud infrastructure for 3 countries, multi-commodity	GreenScope	Pilot success, Phase 2 agreement
6–7	Country selection and prioritization workshop	3 countries prioritized	Joint	EAGC priorities, data availability
7–8	Expanded data integration	Climate (CHIRPS), conflict (ACLED), satellite (Sentinel-2), trade data	GreenScope	Data licensing agreements
8	User research: EAGC analyst workflows	Personas, use cases, interface requirements	Joint	Access to 5+ EAGC staff

2.3.1 Expansion Methodology

- **Country Selection Criteria:** (1) Data richness, (2) Strategic importance to EAGC, (3) Intervention frequency, (4) Regional trade centrality
- **Commodity Coverage:** Maize (priority), beans, wheat, sorghum, rice (if data permits)
- **Corridor Mapping:** Identify 5–7 critical trade routes per country pair (e.g., Kampala-Mombasa, Nakuru-Juba)

2.4 Months 4–6: Module Development and Deployment

Four intelligence modules deployed sequentially:

2.4.1 Module 1: Market Monitoring and Alerting (Month 4)

Component	Function	Data Inputs	Output
Price Anomaly Detection	Flag unusual price movements	RATIN price series, historical volatility	Alert: "Nairobi maize up 15% vs. trend, investigate"
Cross-Market Arbitrage Tracker	Identify profitable trade opportunities	Price pairs, transport costs, border status	Weekly opportunity report
Seasonal Forecasting	3-month price projections	Climate forecasts, planting intentions, stocks	Probabilistic price bands

2.4.2 Module 2: Causal Diagnostics (Month 4–5)

Component	Function	Methodology	Output
Root Cause Analyzer	Attribute price shocks to factors	Causal inference on integrated dataset	Attribution breakdown with confidence
Event Impact Quantifier	Measure specific policy/conflict effects	Synthetic control methods	"Export ban reduced flows by 40%"
Correlation vs. Causation Filter	Distinguish coincidence from causation	Granger causality, instrumental variables	Validated causal relationships

2.4.3 Module 3: Intervention Simulator (Month 5–6)

Component	Function	Scenarios	Output
Reserve Release Optimizer	Timing, quantity, location for reserve deployment	Various release sizes, market conditions	Price impact projection, stock depletion timeline
Trade Facilitation Planner	Corridor investment prioritization	Infrastructure improvements, border efficiency	ROI by corridor, implementation sequencing
Combined Intervention Designer	Multi-tool strategies	Reserves + routes + policy coordination	Synergistic effect estimation
Confidence and Sensitivity Analysis	Robustness of recommendations	Parameter uncertainty, model risk	Confidence intervals, scenario boundaries

2.4.4 Module 4: Strategic Intelligence (Month 6)

Component	Function	Methodology	Output
Pareto Optimization Engine	Resource allocation across countries/interventions	Multi-objective optimization	Efficient frontier: maximum impact per budget
Partnership Readiness Scoring	Assess member state data/intervention capacity	Composite index	Country capability profiles
Donor Briefing Generator	Automated investment case creation	Natural language generation	Customized proposals for AfDB, USAID, etc.

2.5 Months 7–9: Integration and Workflow Embedding

Activity	Description	Deliverable
API Development	RESTful APIs for EAGC systems, member state access	Developer documentation, sandbox environment
Mobile Deployment	Android/iOS apps for field staff	Offline-capable price reporting, alert receipt
Automated Reporting	Scheduled minister briefings, board dashboards	Weekly PDF briefs, real-time web dashboards
EAGC Team Training	Deep-dive for 10+ staff, train-the-trainer	Certified EAGC analysts, training materials
Workflow Integration	Embed into EAGC SOPs for crisis response	Updated standard operating procedures

2.5.1 Training Curriculum

Week	Content
Week 1	Platform fundamentals, data interpretation, navigation
Week 2	Scenario modeling, intervention design, confidence assessment
Week 3	Advanced analytics, model validation, troubleshooting, train-the-trainer
Ongoing	Monthly office hours, quarterly refresher sessions, ad-hoc support

2.6 Months 10–12: Optimization and Learning System

Activity	Description	Target
Intervention Outcome Tracking	Log actual interventions, compare to predictions	Prediction accuracy >80%
Model Retraining	Automated updates with new data, feedback incorporation	Monthly model refresh
User Feedback Loop	Interface improvements, feature requests	NPS >50, feature adoption >70%
Annual Review and Roadmap	Performance assessment, Phase 3 planning	Board presentation, renewed commitment

2.7 Year 1 Success Metrics

Metric	Target	Measurement Method
Decision Speed	75% reduction (3–5 days to <1 day)	Time from question to actionable insight
Intervention Confidence	90% of decisions with quantified confidence	Proportion of recommendations with confidence intervals
Cost Avoidance	Quantified better-targeted interventions	Estimated waste prevented vs. historical baseline
User Adoption	15+ active EAGC users, 5+ member state accesses	Monthly active users, API calls
Model Accuracy	80%+ prediction accuracy for price movements	Backtesting, out-of-sample validation

3 PHASE 3: REGIONAL INTELLIGENCE HUB (Years 2–3)

3.1 Overview

Attribute	Specification
Duration	24 months (scalable thereafter)
Objective	EAGC operates autonomous, Africa-referenceable intelligence infrastructure
Geographic Scope	10+ countries with API access, automated alerts, partnership monetization
Autonomy Level	EAGC owns models and data relationships; GreenScope provides infrastructure layer
Strategic Outcome	EAGC as reference model for African regional economic communities

3.2 Inputs Required from EAGC

Input Category	Specific Items	Format	Owner	Timing
Strategic	10+ country expansion list with prioritization	Board-approved roadmap	EAGC Board	Month 13
	Revenue model and partnership strategy	Business plan, pricing framework	CEO/Commercia	Month 13
	White-label branding requirements	Brand guidelines, approval workflows	Communications	Month 16
	IP ownership and licensing preferences	Legal framework, negotiation mandates	Legal/Board	Month 18
Data	Government MOUs for data sharing (5+ new countries)	Signed agreements, API access	Regional Directors	Months 13–18
	National statistics office partnerships	Data sharing protocols, quality standards	Chief Economist	Months 14–20
	Customs and border agency data feeds	Real-time or near-real-time access	Trade Facilitation	Months 15–24
	Satellite data partnerships (where ground data poor)	Licensing agreements, technical specs	IT/Programs	Months 16–22
Expert Knowledge	Country-specific experts for model calibration (10+ consultants)	Recruitment, contracts, briefings	HR/Regional	Months 13–21
	Academic partnerships for validation	MOUs with universities, CGIAR centers	Research Lead	Months 18–24

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Input Category	Specific Items	Format	Owner	Timing
	Regional economic community relationships (AU, SADC, ECOWAS, IGAD)	Introductions, collaboration frameworks	CEO/Policy	Months 14–24
Operational	EAGC team scale-up (data scientists, analysts, partnership managers)	Job descriptions, hiring, training	HR	Months 13–18
	IT infrastructure handover or continued partnership	Transition plan, SLA agreements	IT Director	Month 24
	Data center or cloud residency decisions	Procurement, compliance review	IT/Legal	Months 16–20
Partnership	Member state subscription agreements	Contract templates, negotiation playbook	Legal/Commercial	Months 18–24
	Development partner project pipeline	Proposal templates, relationship maps	Business Development	Months 16–24
	Commercial trader engagement strategy	Outreach plan, value proposition	Commercial	Months 20–24
Governance	Data ethics committee establishment	Charter, members, meeting schedule	CEO	Month 15
	Scientific advisory board	External expert recruitment, terms of reference	Chief Economist	Month 18
	Technical governance committee (ongoing)	Meeting cadence, decision rights	CTO/IT Director	Quarterly
Decision	Annual roadmap adjustments and investment decisions	Board resolutions	Board/CEO	Quarterly
	Partnership go/no-go decisions	Executive committee approvals	CEO	As needed

3.3 Year 2: Scale and Automate (Months 13–24)

3.3.1 Geographic Expansion (Months 13–18)

Wave	Countries	Selection Criteria	Timeline
2A	Rwanda, Burundi, South Sudan	EAC membership, data partnerships	Months 13–15
2B	Ethiopia, DRC (eastern), Somalia (Somaliland)	Strategic trade corridors, donor interest	Months 16–18
2C	Malawi, Zambia (associate members)	SADC linkage, regional expansion	Months 19–21

Expansion Methodology:

1. **Data Partnership Negotiation:** MOUs with national bureaus of statistics, central banks, customs authorities
2. **Satellite Data Integration:** Where ground data is scarce, use Sentinel-2 for production estimation, night lights for economic activity
3. **Local Calibration:** 4-week country-specific model tuning with local expert input

3.3.2 Automation and Intelligence Maturity (Months 13–24)

Capability	Year 1 State	Year 2 Target	Technology
Data Ingestion	Daily manual/API pull	Real-time streaming	Apache Kafka, webhooks
Model Updates	Monthly retraining	Continuous online learning	Bayesian online learning, drift detection
Anomaly Detection	Rule-based thresholds	AI-powered pattern recognition	LSTM autoencoders, isolation forests
Scenario Generation	Predefined templates	AI-generated "what-if" exploration	GPT-4 class models, simulation trees
Report Generation	Template-based	Fully automated, personalized	NLG, dynamic visualization
Quality Assurance	Human review	Automated validation + spot checks	Statistical process control

3.3.3 Autonomous Operations Architecture

```
[Data Layer]      → [Real-time Processing] → [Causal Engine]    → [Decision Layer] → [Distrib
     ↓                      ↓                       ↓                     ↓                ↓
Multiple sources    Stream processing      Automated inference   Alert generation   APIs, app
(10+ countries)     (Apache Flink)         (Continuous)          (Threshold-based)  (Multi-ch
```

3.4 Year 3: Ecosystem and Monetization (Months 25–36)

3.4.1 Partnership Infrastructure (Months 25–30)

Partnership Type	Offering	Input Required from EAGC	Target Partners
Member State Subscriptions	Country-specific dashboards, early warning	Pricing approval, contract negotiation, onboarding support	Agriculture ministries, central banks
Development Partner Analytics	Custom analysis, impact evaluation	Proposal development, relationship management, reporting	AfDB, USAID, FAO, WFP
Commercial Trader APIs	Real-time arbitrage signals, route optimization	Commercial strategy, legal frameworks, customer success	Regional grain traders, logistics firms
Academic/Research Access	Anonymized datasets, methodology papers	Data governance, ethical review, publication rights	Universities, CGIAR centers

3.4.2 White-Label and Licensing (Months 28–36)

- **EAGC-branded Platform:** "EAGC Intelligence" powered by GreenScope
- **IP Ownership Transition:** EAGC owns models trained on their data; GreenScope provides infrastructure
- **Replication Licensing:** Other regional bodies (SADC, ECOWAS, IGAD) license the architecture

3.4.3 Advanced Capabilities (Months 25–36)

Capability	Description	Strategic Value	Input Required
Predictive Trade Policy	Forecast member state export restriction likelihood	Early diplomatic intervention	Historical policy database, diplomatic cables
Climate-Conflict Nexus Modeling	Anticipate compound shocks	Pre-positioning, resilience planning	Conflict data partnerships, climate projections
Smallholder Integration	Farmer SMS alerts, voice-based price reporting	Inclusive market participation	Mobile network partnerships, farmer organization contacts
Blockchain Traceability	Origin verification for premium markets	Quality differentiation, export readiness	Technology partnerships, certification body agreements
Pan-African Dashboard	Continental view with EAGC as node	AU engagement, policy influence	AU relationship, other regional body data sharing

3.5 Governance and Sustainability (Ongoing)

Element	Structure	Responsibility	Input Required
Technical Governance	Quarterly architecture board	GreenScope CTO + EAGC IT Director	Meeting time, decision authority, budget oversight
Data Governance	Data ethics committee	EAGC Legal + member state reps	Committee charter, member appointments, case review
Scientific Advisory	Annual external review	Independent academics, practitioners	Expert recruitment, review scope, implementation
Business Model	Cost recovery + surplus for R&D	EAGC Finance + GreenScope	Financial targets, investment decisions, audit

4 CROSS-CUTTING METHODOLOGY: QUALITY ASSURANCE

4.1 Quality Assurance at Every Phase

QA Layer	Method	Frequency	Input Required
Data Quality	Automated validation rules, anomaly flagging, manual spot checks	Continuous	Data quality thresholds, escalation protocols
Model Validation	Backtesting, cross-validation, expert review, out-of-sample testing	Monthly (Phase 2+), end of Phase 1	Expert time for review, historical outcome data
User Acceptance	Usability testing, task completion rates, satisfaction surveys	End of each module deployment	User time, feedback incorporation process
Outcome Tracking	Intervention logging, prediction vs. actual comparison, bias detection	Quarterly	Intervention records, outcome data, feedback loops
Security and Compliance	Penetration testing, data residency audits, access log review	Bi-annually	Security policies, audit access, remediation resources

5 REALISTIC TIMELINE AND RISK SUMMARY

5.1 Phase Summary

Phase	Duration	Critical Path Risk	Mitigation	Key Input for Success
Pilot	3 weeks	Data access delays	Parallel workstreams, proxy data ready	CEO sponsorship, economist time, clear question
Platform	10 months	User adoption	Embedded training, workflow integration	Dedicated counterpart, IT integration support, country partnerships
Hub	24 months	Country data partnerships	Government relations, satellite fallback	Board-approved expansion list, government MOUs, team scale-up

5.2 Key Dependencies Across All Phases

1. **EAGC CEO sponsorship** — Essential for resource allocation and priority setting
2. **Dedicated EAGC counterpart time** — 20% (Pilot), 20% (Platform), 50%+ (Hub)
3. **Data partnership negotiations** — Government relations capacity at EAGC
4. **IT security and integration** — EAGC IT team engagement and approval
5. **Board and member state buy-in** — Strategic alignment and investment commitment